

Advanced SQLExercise 6Date Functions

Table 1: Employees

emp_id	name	hire_date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Q1. Add 6 months to each employee's hire date using DATEADD().

SQL Query

```
SELECT emp_id,
```

```
name,
```

```
hire_date,
```

```
DATEADD(MONTH, 6, hire_date) AS hire_plus_6_months
```

```
FROM Employees;
```

Output Results

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

Student_id	name	dob
101	Maya	2005-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

Q2. Use DATEIFF() to find age in days from dob to today

SQL Query

```
SELECT student_id,
       name,
       DATEIFF(DAT, dob, '2026-04-22') AS age-in-days
FROM Students;
```

Output Results

Student_id	name	age-in-days
101	Maya	7560
102	Ethan	7812
103	Sienna	7343

Table 3: Events

event_id	event-name	event-date
1	Seminar	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

Q3. Find how many days are left until each event using DATEIFL.

SQL Query

```
SELECT event_id,  
       event_name,  
       DATEIFL(DAT '2026-04-22', event_date) AS days_remaining  
FROM Events;
```

Output Results

event_id	event_name	days_remaining
1	Seminar	-677
2	Workshop	-233
3	Hackathon	-457

Table 4: Invoices

event-id	event-name	event-date

Table 4: Invoices

invoice-id	issue-date	due-date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

Q4: Calculate the number of days between issue-date and due-date.

SQL Query

SELECT invoice-id,
issue-date,
due-date,

DATEIF(DAT, issue-date, due-date) AS days-between

FROM Invoices;

Output Results

invoice-id	issue-date	due-date	days-between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5: Courses

Course-id	name	start-date
201	SQL Basics	2025-05-01
202	Python	2025-06-01

Q5. Format start-date as 'Month YYYY' using TO_CHAR().

SQL Query

SELECT course_id,

name,

TO_CHAR(start-date, 'Month YYYY') AS formatted_date

FROM Courses;

Output Results

Course-id	name	formatted-date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6: Memberships

member-id	start-year	start-month	start-day
11	2023	5	10
12	2022	11	25

Q6. Create full date from parts using DATE_FROM_PARTS().

SQL Query

SELECT member_id

DATE_FROM_PARTS(start-year, start-month, start-day) AS full_start-date

FROM Memberships;

Output Results

member_id	Full-start date
1	2023-05-10
2	2022-11-25

Table 1: Subscriptions

Sub-id	Plan	renewal date
11	Basic	2025-01-01
12	Premium	2025-03-15

Q7. Extend each renewal-date by 1 year using DATEADD.

SQL Query

```
SELECT sub-id,  
       plan,  
       DATEADD(YEAR, 1, renewal-date) AS extended-renewal-date  
FROM Subscriptions;
```

Output Results

Sub-id	Plan	extended-renewal-date
11	Basic	2026-01-01
12	Premium	2026-03-15

Table 8: Orders

Order-id	Order-date
1001	2025-04-15
1002	2025-04-10

Q8. Show current date and difference from order-date. Use CURRENT_DATE and DATEDIFF.

SQL Query

```
SELECT order-id,
       order-date,
       '2026-04-22' AS today-date,
       DATEDIFF(DAT, order-date, '2026-04-22') AS days-since-order
FROM Orders;
```

Output Results

Order-id	order-date	today-date	days-since-order
1001	2025-04-15	2026-04-22	372
1002	2025-04-10	2026-04-22	377

Table 9: Trainings

training-id	topic	training-date
1	Safety	2025-01-10
2	Compliance	2025-02-20

Q9. Extract the year from training-date using DATE_PART() or EXTRACT().

SQL Query

```
SELECT training-id,
       topic,
       EXTRACT (YEAR FROM training-date) AS training-year
FROM Trainings;
```


Output Results

training-id	topic	training-year
1	Safety	2025
2	Compliance	2025

Table 10: Blog-Posts

post-id	title	published-on
1	SAL Tips	2025-04-01 10:15:00
2	Data Cleaning	2025-04-12 16:45:00

Q10. Extract hour and minute from published-on.

SQL Query

```
SELECT post-id,  
       title,  
       EXTRACT (Hour FROM published-on) AS hour-published,  
       EXTRACT (Minute FROM published-on) AS minute-published  
FROM Blog-Posts;
```

Output Results

Post-id	title-id	hour-published	minute-published
1	SAL Tips	10	15
2	Data Cleaning	16	45

Table 11: Drivers

driver-id	license-expiry
301	2025-08-10
302	2023-12-31

Q11. Calculate days left until license expiry using DATEDIFF() and today's date.

SQL Query

```
SELECT driver-id,
       license-expiry,
       DATEDIFF(DATETIME, '2026-04-22', license-expiry) AS days-left
FROM Drivers;
```

Output Results

driver-id	license-expiry	days-left
301	2025-08-10	- 255
302	2023-12-31	- 843

Table 12: Messages

Message-id	Sent-timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

Q12. Display the current timestamp and calculate seconds since the message was sent.

SQL Query

```
SELECT message-id,
       Sent-timestamp,
       '2026-04-22 00:00:00' AS current-timestamp,
       DATEDIFF(SECOND, Sent-timestamp, '2026-04-22 00:00:00') AS seconds-since-sent
FROM Messages;
```


Output Results

message_id	Sent_timestamp	Current_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	2026-04-22 00:00:00	31789635
2	2025-04-18 23:59:59	2026-04-22 00:00:00	31881601

Table 13: Returns

return_id	return_date
901	2025- 03-01 ⁰⁴⁻⁰⁵
902	2025- 03-05 ⁰⁴⁻⁰¹

Q13. Add 15 days to return_date using DATEADD() to show restock_date.

SQL Query

```
SELECT return_id  
       return_date  
       DATEADD(DAY, 15, return_date) AS restock_date  
FROM Returns;
```

Output Results

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 4: Projects

Project ID	Project Name
1	Project A
2	Project B

SQL Query to get Project ID and Project Name from Table Projects

SQL Query

SELECT Project ID

FROM Projects

Table 5: Meetings

Output results

Project ID	Assigned On Date
1	2025-03-01
2	2025-03-05

Table 6: Meetings

Meeting ID	Scheduled Time
1	2025-03-01 14:00:00
2	2025-03-01 17:00:00

SQL Query to get Meeting ID and Scheduled Time from Table Meetings

SQL Query

SELECT Meeting ID

FROM Meetings

Table 7: Meetings

Output Results

meeting_id	formatted_meeting_time
1	April 19, 2025 at 02:00PM
2	April 19, 2025 at 09:30AM